STUDENT PERFORMANCE ANALYTICS SYSTEM

Project Documentation

Python with AI | Student Performance Data Analysis

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GitHub Repository: https://github.com/jayasreebalaji14/student-performance-analytics-system

1. Project Title

Student Performance Analytics System

A Python-based tool that processes marks for 15 students and reports individual grades, pass/fail results, and class-level performance statistics.

2. Overview

- Command-line Python project using pandas and NumPy.
- Dataset: 15 students (IDs 501–515), five subjects marked out of 100, and attendance percentages.
- Processing flow: read data → calculate total and average → assign grade → determine pass/fail → analyze class results.
- The supplied data includes a raw student dataset and a processed performance report.

Technology Stack

Component	Purpose
Python 3	Core programming language and project logic
pandas	Reading and organizing CSV data, calculating values, and preparing reports
NumPy	Numerical operations and rounding average values
CSV files	Storing raw student records and processed performance results
GitHub	Version control and online project submission

Project Structure

File	Description
main.py	Main script containing the processing and analysis logic
students.csv	Raw student records, as shown in the uploaded dataset
student_performance.csv	Processed records with Total, Average, Grade, and Result
README.md	Project overview and usage notes
Project_Documentation.docx / .pdf	Description of the project, implementation, results, and conclusion

Note: The repository screenshot showed both students.csv and student_performance.csv. Confirm that main.py reads and writes the intended filenames.

3. Objectives

1. Create a structured dataset containing student IDs, names, five subject marks, and attendance.
2. Store and read student records using CSV files.
3. Calculate each student's total marks out of 500 and average marks.
4. Assign a letter grade based on the average.
5. Determine pass/fail using a minimum mark of 35 in every subject.

6. Calculate class-level average, highest and lowest averages, and pass/fail counts.
7. Save the processed performance report to a CSV file.

4. Implementation

4.1 Data and Storage

The project dataset contains 15 records with the columns Student_ID, Name, Python, SQL, HTML, CSS, JavaScript, and Attendance. The processed report adds Total, Average, Grade, and Result.

4.2 Total and Average Calculation

Total is the sum of the five subject marks. Average is the total divided by five, rounded to one decimal place in the supplied report.

Example: Priyanka scored 92, 92, 94, 88, and 98. Her total is 464 and her average is 92.8.

4.3 Grade Assignment

Average	Grade
90 and above	A+
80–89.9	A
70–79.9	B
60–69.9	C
50–59.9	D
40–49.9	E
Below 40	F

4.4 Pass/Fail Logic

A student passes only when every subject mark is 35 or more. If any subject mark is below 35, the student fails regardless of the average. Attendance is recorded in the dataset but does not affect the reported pass/fail result.

4.5 Class Analysis

The report summarizes the class average, highest and lowest student averages, pass/fail counts, and grade distribution. Subject-wise averages are also calculated below from the supplied data.

4.6 Report Export

The processed report contains the original student columns plus Total, Average, Grade, and Result. The file shown in the supplied repository is student_performance.csv.

4.7 How to Run

Install the required libraries if they are not already installed:

```
pip install pandas numpy
python main.py
```

Run these commands in the project folder. Confirm that the filenames in main.py match the CSV files in the repository.

5. Final Outcome

5.1 Class Analysis Results

Metric	Value
Number of students	15
Class average (mean of student averages)	79.99
Highest average	92.8 (Priyanka, ID 503)

Lowest average	47.8 (Jas, ID 515)
Students passed	14
Students failed	1

5.2 Grade Distribution

Grade	Students	Count
A+	Priyanka, Krishna, Harshini, Suneetha	4
A	Jaya sree, Vijay, Balaji, Sarika, Radha	5
B	Priya, Diana, Jack Daneil	3
C	Pavan Kumar, Sha	2
D	None	0
E	Jas	1
F	None	0

5.3 Subject-wise Average

Subject	Average mark (out of 100)
Python	76.87
SQL	80.67
HTML	83.53
CSS	77.60
JavaScript	81.27

5.4 Key Observations

- Top performer: Priyanka (ID 503) with total 464, average 92.8, and grade A+.
- Lowest average: Jas (ID 515) with total 239, average 47.8, and grade E.
- 14 students passed and 1 student failed under the minimum-35-marks-per-subject rule.
- 9 students earned an A or A+ grade.
- Attendance is included in the dataset but is not used to determine grade or pass/fail status in the supplied results.

5.5 Deliverables

- Python script (main.py).
- Raw student dataset (students.csv).
- Processed performance report (student_performance.csv).
- README.md with project overview and instructions.
- This project documentation file.

5.6 Limitations and Future Enhancements

- Validate CSV filenames and required columns before processing.
- Add an optional attendance threshold if attendance should affect eligibility.
- Add rankings and subject-wise topper lists.
- Generate charts to make performance trends easier to understand.

5.7 Conclusion

The Student Performance Analytics System automates marks processing, grade assignment, pass/fail classification, and summary statistics for a class of 15 students. It demonstrates practical use of Python, pandas, NumPy, and CSV files to turn student records into a clear performance report.